Proof. Suppose I is invertible, with $IJ = A$. Then $jI \subseteq A$ for all $j \in J$, so $J \subseteq A \div I$, and $A = IJ \subseteq I(A \div I) \subseteq A$, so $I(A \div I) = A$. The reverse implication is immediate. $\square$

In the next lecture we will prove that in a Dedekind domain every nonzero fractional ideal is invertible, but let us first note that this is not true in general.

Example 2.21. Consider the subring $A := \mathbb{Z}+2i\mathbb{Z}$ of the Gaussian integers (with $i^2 = -1$). The set $I := 2\mathbb{Z}[i]$ is a non-invertible A-ideal (even though it is an invertible $\mathbb{Z}[i]$-ideal); indeed, we have $A \div I = \mathbb{Z}[i]$ and $I(A \div I) = 2\mathbb{Z}[i] \subsetneq A$.

2.6 Invertible fractional ideals and the ideal class group

In this section A is a noetherian domain (not necessarily a Dedekind domain) and K is its fraction field. Recall that a fractional ideal of A is a finitely generated A-submodule of K, and if I and J are fractional ideals, so is the ideal quotient

$$I \div J := \{x \in K : xJ \subseteq I\},$$

and we say that a fractional ideal I is invertible if $IJ = A$ for some fractional ideal J. The definition of $A \div I$ implies $I(A \div I) \subseteq A$, and Lemma 2.20 implies that I is invertible precisely when this inclusion is an equality, in which case the inverse of I is $A \div I$.

Ideal multiplication is commutative and associative, thus the set of nonzero fractional ideals of a noetherian domain form an abelian monoid under multiplication with $A = (1)$ as the identity. It follows that the subset of invertible fractional ideals is an abelian group.

Definition 2.22. The *ideal group* $\mathcal{I}_A$ of a noetherian domain A is the group of invertible fractional ideals. Note that, despite the name, elements of $\mathcal{I}_A$ need not be ideals.

Every nonzero principal fractional ideal (x) is invertible (since $(x)^{-1} = (x^{-1})$), and a product of principal fractional ideals is principal (since $(x)(y) = (xy)$), as is the unit ideal (1), thus the set of nonzero principal fractional ideals $\mathcal{P}_A$ is a subgroup of $\mathcal{I}_A$.

Definition 2.23. Let A be a noetherian domain. The quotient $\mathrm{cl}(A) := \mathcal{I}_A/\mathcal{P}_A$ is the *ideal class group* of A; it is also called the *Picard group* of A and denoted $\mathrm{Pic}(A)$.[3]

Example 2.24. If A is a DVR with uniformizer π then its nonzero fractional ideals are the principal fractional ideals (π^n) with $n \in \mathbb{Z}$ (including $n \le 0$). We have $(\pi^m)(\pi^n) = (\pi^{m+n})$, thus the ideal group of A is isomorphic to $\mathbb{Z}$ (under addition). In this case $\mathcal{P}_A = \mathcal{I}_A$ and the ideal class group $\mathrm{cl}(A)$ is trivial.

Remark 2.25. A Dedekind domain is a UFD if and only if its ideal class group is trivial (we will prove this in the next lecture), thus $\mathrm{cl}(A)$ may be viewed as a measure of how far A is from being a UFD.

[3] In general, the Picard group of a commutative ring A as the group of isomorphism classes of A-modules that are invertible under tensor product (equivalently, projective modules of rank one). When A is a noetherian domain, the Picard group of A is canonically isomorphic to the ideal class group of A and the two notions may be used interchangeably.